TECHNICAL SPECIFICATION FOR ALL ALUMINIUM ALLOY CONDUCTOR - PANTHER (AAAC)

TABLE OF CONTENTS

OF

ALL ALUMINIUM ALLOY CONDUCTOR

Sl. No	**DESCRIPTION**	**PAGE NO.**
1.0	Scope	3
2.0	Standards	3
3.0	General	4
4.0	Physical constants for aluminium alloy wires	4
5.0	Standard sizes	5
6.0	Joints in wires	6
7.0	Stranding	7
8.0	Lengths & variations in length	7
9.0	Tests	7
10.0	Rejection & retests	10
11.0	GTP	77

TECHNICAL SPECIFICATION FOR ALL ALUMINIUM ALLOY CONDUCTOR

1. 0 SCOPE

This specification covers design, Engineering, Manufacture, Testing, Inspection before dispatch, forwarding, packing, transportation to sites, Insurance (both during transit & storage), storage, erection, supervision testing & commissioning of all sizes of All Aluminum Alloy Conductors of the Aluminum–Magnesium-Silicon type for use in the distribution overhead power lines.

The conductor offered shall be procured from short listed vendor at **E-23** and shall have been successfully Type Tested during last five years on the date of bid opening. The Type Test reports shall be submitted along with the bid.

The AAA Conductor shall conform in all respects to highest standards of engineering, design, workmanship complying this specification and the latest revisions of relevant standards at the time of offer and the Owner shall have the power to reject any work or materials, which, in his judgement, is not in full accordance therewith.

2.0 STANDARDS

Except where modified by the specification, the Aluminum Alloy Conductor shall be designed, manufactured and tested in accordance with latest editions of the following standards:

Sl.No	International Standard	IS	Description
1	IEC :61089		Round wire concentric lay overhead electrical standard Conductor
2		IS 398 part-4	Aluminum Alloy Stranded Conductors
3		IS 9997	Aluminum Alloy redraw rods for electrical purposes
4	IEC 60104		Aluminum Magnesium Silicon alloy wire for overhead line conductors
5		IS 1778	Reels and drums of bare conductor.

3.0 GENERAL

The wires shall be of heat treated aluminium, magnesium silicon alloy containing approximately silcon-0.5 to 0.9 %, magnesium-0.6 % to 0.9%, Fe- 0.5% (maximum), Copper- 0.1% (max), Mn- 0.03%, Cr-0.03%, Zn-0.1%, B- 0.06%, others (each) 0.03% others (Total) 0.1 % (max) Aluminium reminder and having the mechanical and electrical properties specified in the table and be smooth and free from all imperfections, such as, spills, splits and scratches.

Neutral grease shall be applied between the layers of wires. The drop point temperature of the grease shall not be less than 120^0C.

3.1 Mechanical and Electrical Characteristics of Aluminium Alloy Wires used in Construction of Stranded Aluminium Alloy Conductors

Nominal Diameter	Minimum Diameter	Max. Diameter	Cross Sectional Area	Mass	Minimum Breaking Load		Maximum Resistance at 20^0 C
					Before stranding	After stranding	
1	2	3	4	5	6	7	8
mm	mm	mm	mm^2	Kg/km	KN	KN	ohms/km
2.88	2.85	2.91	6.514	17.59	2.02	1.92	5.141

Maximum resistance values given in column 8 have been calculated from the maximum values of the resistivity as specified and the cross sectional area based on the minimum diameter.

The minimum breaking load is calculated on nominal diameter at ultimate tensile strength of 67.49 KN for wire before stranding and 95% of the ultimate tensile strength after stranding.

4.0 PHYSICAL CONSTANTS FOR ALUMINIUM ALLOY WIRES

4.1 Resistively:

For the purpose of this specification, the standard value of resistivity of aluminum alloy wire which shall be used for calculation is to be taken as 0.0325 ohm-mm^2/m at 20^0 C. the maximum value of resistivity of any single wire shall not, however, exceed 0.0328 ohm-mm^2/m at 20^0 C.

4.2 Density:

At a temperature of 20^0 C, the density of aluminum alloy wire is to be taken as 2700 kg/m^3.

4.3 Temperature Coefficient of Linear Expansion:

The temperature coefficient of linear expansion of aluminium alloy wire is to be taken as 23 x 10^{-6} /^{0}C.

4.4 Constant – Mass Temperature Coefficient

At a Temperature of 20^0 C, the constant – mass temperature coefficient of resistance of aluminium alloy wires, measured between two potential points rigidly fixed to the wire, is taken as 0.00360/^{0} C.

5.0 STANDARD SIZES

5.1 Nominal Sizes of Wires

The aluminium alloy wires for standard constructions covered by this specification shall have the diameters as specified in the table and a tolerance of ±1% shall be permitted on the nominal diameter.

5.2 Standard Conductors

The sizes, resistance and masses (excluding the mass of grease) of stranded aluminium alloy conductors shall be as given in table. The preferred sizes are highlighted in the table.

5.3 Mechanical and Electrical Characteristics of Aluminium Alloy Stranded Conductors

Sl. No.	Actual Area	Stranding and Wire Dia	Approx. Overall Dia	Approx. Mass	Calculated Maximum Resistance at 20^0 C	Approx Calculated Breaking
1	2	3	4	5	6	7
	mm^2	mm	mm	kg/km	ohms/km	KN
2	241.0	37/2.88	20.16	663.8	0.1417	67.49

5.3.1 Increase in Length due to Stranding

When straightened out, each wire in any particular layer of a stranded conductor, except the central wire, is longer than the stranded conductor by an amount depending on the lay ratio of that layer.

5.3.2 Resistance and Mass of Conductor

The resistance of any length of stranded conductor is the resistance of the same length of any one wire multiplied by a constant as set out in the table below.

The mass of each wire in any particular layer of the stranded conductor, except the central wire, will be greater than that of an equal length of straight wire by an amount depending on the lay ratio of that layer.

The total mass of any length of an aluminium stranded conductor is, therefore, obtained by multiplying the mass of an equal length of straight wire by an appropriate constant as mentioned below.

In calculating the stranding constants as mentioned in the table below, the mean ratio, that is the arithmetic mean of the relevant minimum and maximum values in table for lay ratio has been assumed for each layer.

5.3.3 Calculated Breaking Load of Conductor

For a conductor containing **not more than** 37 wires, 95% of the sum of strength of the individual wires calculated from the values of the minimum breaking load given in this specification.

For a conductor containing **more than** 37 wires, 90% of the sum of the strengths of the individual wire calculated from the values of the minimum breaking load given in this specification.

5.3.4 Calculated Area and Maximum Resistance of Conductor

The actual area of a stranded conductor has been taken as the sum of the cross-sectional areas of the individual wires of nominal diameter.

Maximum resistance values of stranded conductor have been calculated on the basis of maximum resistivity and the cross-sectional area based on the minimum diameter of wires.

5.4 Stranding Constants

Number of Wires in Conductor	**Stranding Constants**	
	Mass	Electrical Resistance
(1)	(2)	(3)
37	37.74	0.02757

6.0 JOINTS IN WIRES

6.1 Conductor containing seven wires

There shall be no joint in any wire of a stranded conductor containing seven wires, except those made in the base rod or wire before final drawing.

6.2 Conductors containing more than seven wires

In stranded conductors containing more than seven wires, joints in individual wires are permitted in any layer except the outermost layer (in addition to those made in the base rod or wire before final drawing) but no two such joints shall be less than 15 m apart in the complete stranded conductor. Such joints shall be made by cold pressure butt welding. They are not required to fulfill the mechanical requirements for un-jointed wires.

7.0 STRANDING

The wire used in the construction of a stranded conductor shall, before and after stranding, satisfy all the relevant requirements of this standard.

The lay ratio of the different layers shall be within the limits given in the table for lay ratio.

In all constructions, the successive layers shall have opposite directions of lay, the outermost layer being righ-handed. The wires in each layer shall be evenly and closely stranded.

In aluminium alloy stranded conductors having multiple layers of wires, the lay ratio of any layer shall not be greater than the lay ratio of the layer immediately beneath it.

7.1 Lay Ratios for Aluminium Alloy Stranded Conductors

Number of Wires in Conductor	LAY RATIOS							
	3/6 Wire Layer		12 Wire Layer		18 Wire Layer		24 Wire Layer	
	Min	Max	Min	Max	Min	Max	Min	Max
37	10	14	10	13	10	12	---	---

NOTE:For the purpose of calculation the mean lay ratio shall be taken as the arithmetic mean of the relevant minimum and maximum values given in this table

8.0 LENGTHS AND VARIATIONS IN LENGTHS:

Unless otherwise agreed between the Owner and the Contractor, stranded aluminium alloy conductors shall be supplied in the manufacturer's usual production lengths to be indicated in the bid Schedule. The Owner reserves the right to specify particular lengths of conductor such that certain drum lengths will be shorter than others. There will in both cases be a permitted variation of –0 + 5% in the length of any one conductor length.

9.0 TESTS

9.1 Type Tests

The following tests should have been carried out as per relevant ISS.

9.1.1 Ultimate Tensile Strength Test

This test is intended to confirm not only the breaking strength of the finished conductor but also that the conductor has been uniformly stranded.

A conductor sample of minimum 5mtr. length fitted with compression dead end clamps at either end shall be mounted in a suitable tensile test machine. Circles perpendicular to the axis of the conductor shall be marked at two places on its surface. Tension on the conductor sample shall be increased at a steady rate up to 50% of the minimum UTS specified and held for one minute. The circles drawn shall not be distorted due to relative movement of the individual strands. Thereafter the load shall be increased at a steady rate to the specified minimum UTS and held at that load for one minute. The conductor sample shall not fail during this period. The applied load shall then be increased until the failing load is reached and the value recorded.

9.1.2 D.C Resistance Test

On a conductor sample of minimum 5mtr. length two contact clamps shall be fitted with a predetermined bolt torque. The resistance between the clamps shall be measured using a Kelvin double bridge by initially placing the clamps at zero separation and subsequently one meter apart. The test shall be repeated at least five times and the average value recorded. The value obtained shall be corrected to the value at 20^0 C, which shall conform to the requirements of this specification.

9.2 Routine Tests

9.2.1 Measurement of Physical Dimensions:

The samples should meet the desired dimensional requirements before conducting following Routine Tests as per relevant ISS.

9.2.2 Selection of Test Samples

Samples for the tests specified in this specification shall be taken by the manufacturer before stranding, from not less than 10% of the individual lengths of aluminium alloy wire included in any one final heat-treatment batch and which will be included in any one consignment of the stranded conductors to be supplied.

Samples shall then be obtained by cutting 1.2 meters from the outer end of the finished conductor from not more than 10% of the finished reels or drums.

Tests for electrical and mechanical properties of aluminium alloy wire shall ordinarily be made before stranding since wires unlaid from conductors may have different physical properties from those of the wire prior to stranding because of the deformation brought about by stranding and by straightening for test. Spools offered for inspection shall be divided into equal lots, the number of lots being equal to the number of samples to be selected, a fraction of a lot being counted as s complete lot. One sample spool shall be selected at random from each lot.

9.3 Breaking Load Test

The breaking load of one specimen, cut from each of the samples taken shall be determined by means of a suitable tensile testing machine. The load shall be applied gradually and the rate of separation of the jaws of the testing machine shall be not less than 25 mm / min and not greater than 100mm /min.

9.4 Elongation Test

The elongation of one specimen cut from each of the samples taken shall be determined as follows :

The specimen shall be straightened by hand and an original gauge length of 200 mm shall be marked on the wire. A tensile load shall be applied as described above and the elongation shall be measured after the fractured ends have been fitted together. If the fracture occurs outside the gauge marks, or within 25 mm of either mark, and the required elongation is not obtained, the test shall be disregarded and another test should be made.

When tested before and after stranding, the elongation shall not be less than 4% on a gauge length of 200 mm.

9.5 D.C Resistance Test

The electrical resistance test of one specimen cut from each of the samples taken shall be measured at ambient temperature. The measured resistance shall be corrected to the value at 20^0 C by means of the formula

$R_{20} = R_T [1/(1+\alpha(T-20)]$

Where R_{20} = Resistance corrected at 20°C

R_T = Resistance measured at T^0C

A = Constant – Mass temperature coefficient of resistance, 0.0036, and

T = ambient temperature during measurement

The resistance corrected at 20^0 C shall not be more than the maximum values specified.

9.6 Chemical Analysis of Aluminium Alloy

Samples taken from the alloy coils / strands shall be chemically / spectrographically analysed. The results shall conform to the requirements stated in this specification. The Contractor shall make available material analyses, control documents and certificates from each batch as and when required by the **APTRANSCO**.

Test should be conducted at the independent test house by the Owner in the case of absence of facility at manufacturer. However the cost of such testing shall be borne by the manufacturer/Contractor.

9.7 Dimensional and Lay Length Check

The individual strands of the conductors shall be dimensionally checked and the lay lengths checked to ensure that they conform to the requirements of this specification.

Ten percent drums from each lot shall be rewound in the presence of the Owner or his representative to allow visual checking of the conductor for joints, scratches or other surface imperfections and to ensure that the conductor generally conforms to the requirements this specification. The length of conductor would on the drum shall be re-measured by means of an approved counter / meter during the rewinding process.

9.8 Visual and dimensional Checks on the Conductor Drums.

9.8.1 The drums shall be visually and dimensionally checked to ensure that they conform to the requirements of this specification and of IS 1778: Specification for reels and drums of bare conductors. For wooden drums, a suitable barrel batten strength test procedure is required. The Bidder shall state in his bid the tests to be carried out on the drums and shall include those tests in the Quality Assurance Programme.

9.8.2 Acceptance Tests :

The following acceptance tests as per IS-398 (PartIV)/1994 with latest amendments if any, shall be carried out on all samples

i. Measurement of lay ratio

ii. Measurement of diameters of individual wire

iii. Measurement of resistance of individual wire

iv. Breaking load test of individual wire

v. Elongation test of individual wire

vi. Torsional test of individual wire

10.0 REJECTION AND RETESTS

10.1 Type Tests

Should the conductor fail any of the type tests specified above, the Owner will not accept any conductor manufactured from the material, nor conductor made by the manufacturing methods used for the conductor which failed the test. The manufacturer shall propose suitable modifications to his materials and techniques in order that he can produce conductor which will satisfactorily pass the type test requirements.

10.2 Routine Tests

Should any one of the test pieces first selected fail the requirements of the tests, two further samples from the same batch shall be selected for testing, one of which shall be from the length from which the original test sample was taken unless that length has been withdrawn by the manufacturer.

Should the test pieces from both these additional samples satisfy the requirements of the tests, the batch represented by these samples shall be deemed to comply with the standard. Should the test pieces from either of the two additional samples fail, the batch represented shall be deemed not to comply with the standard.

If checked on individual strand diameters, conductor lay lengths and conductor surface condition indicate non-compliance with the requirements of the specification, the particular drum will be rejected. Inspection will then be carried out on two further drums within the same batch. If the conductor on either of the drums is non-complaint, the complete batch will be rejected.

10.3 Delivery Extension due to Rejection of Conductor

The rejection of conductor due to its failure to pass either type or routine tests shall not permit the Contractor to apply for any extension to the time period within which he has contracted to complete the Project.

11.0 GUARANTEED TECHNICAL PARTICULARS:

GTP/DATA SHEET FOR AAA PANTHER CONDUCTOR Reference Standard IS 398 PART-4(1994)		
SR. NO	**DESCRIPTION**	**PARTICULARS**
1	Standard according to which the conductor will be manufactured and tested	IS 398 PART-4(1994)
	The type-tests should have been conducted not earlier than 5 years in the Standard third party laboratory. The Manufacturer shall produce the type test reports at the time of acceptance tests.	
2	Quality of material & standard to which conform	BIS
	The Manufacturer shall produce the valid BIS certification at the acceptance tests.	
3	Number of Strands (Aluminium Alloy)	Alu. Alloy 37/2.88mm
4	**I) Diameter of Stands (Aluminium Alloy)**	
	a) Nominal	2.88
	b) Maximum	2.91
	c) Minimum	2.85
	II) Overall Diameter of conductor	20.16
5	Cross sectional Area of	
	a) Whole Conductor	241.00
	b) Each Strand (At Nominal Diameter)	6.514
6	Minimum % of Aluminium in Alloy	Remainder IS 398 Part-4(1994) References IS 9997 : 1981
7	**Number of Strands**	
	a) Center wire (Aluminium in Alloy)	1
	b) First Layer(Aluminium in Alloy)	6
	c) Second Layer (Aluminium in Alloy)	12
	d) Third Layer	18
8	Weight of	
	a) Whole Conductor	663.8
	b) Aluminium Alloy Strand (At Nominal Diameter)	17.59
9	**Calculated D.C Resistance at 20°C**	
	a) Whole Conductor	0.1417
	b) Aluminium Alloy Strands	5.141
10	Ultimate Tensile strength	
	a) Whole Conductor	67.49 (Minimum)
	b) Minimum Breaking Load before Stranding (Aluminium Alloy Strands)	2.02 (Minimum)
	c) Minimum Breaking Load after Stranding (Aluminium Alloy Strands)	1.92 (Minimum)
11	**Modulus of elasticity of**	
	Aluminium Alloy strands	
	a) Whole Conductor	0.5814×10^6

12	**Co-efficient of linear expansion per degree Centigrade of**		
	Aluminium Alloy strands	23.0 x 10^{-6}	
	a) Whole Conductor	23.0 x 10^{-6}	
13	Resistivity (Ohms.mm^2/m 20° C)	0.0328	
14	a) Continous maximum current rating of conductor in still air at ambient Temperature (40° C)	@ 85°C	
	b) Temperature rise for the above on °C	45°C	
15	Lay Ratios	Minimum	Maximum
	a) Aluminium Alloy Strands : 1st Layer (6 wires)	12	14
	b) Aluminium Alloy Strands : 2nd Layer (12wires)	11	13
	c) Aluminium Alloy Strands : 3rd Layer (18 wires)	10	12
16	Maximum working tension	70% of UTS of Conductor	
17	Maximum single length of conductor which can be manufacture	1.5	
18	Tolerance, if any, on standard length.	5 ±%	
19	No. of standard length in one reel	1	
20	Weight of the conductor in one reel	995.7	
21	Weight of the drum	250 kg ± 5 %	
22	Gross weight of the real including weight on the conductor	1245.7	
23	Drum(Reel) Details	1220mm x 600mm x 710 mm	
	a) Dimensions of the Reel	IS 1778/1980	
	b) Whether the drum on which the conductor is wound conforms to the specification	IS 1778/1980	
24	Other particulars, if any	Brand Name other details to be provided on the surface of drum.	
25	Raw materials are to be procured directly from the primary Produces so as to ensure the quality of raw materials. The test certificates of raw materials and invoices shall be produced at the time of inspection.		
26	a) Aluminium Alloy Rods	Aluminium Rods	
		1) Hind Aluminium Ind. Ltd.	
		2) Associated Aluminium Pvt.Ltd	
	b) Aluminium ingot	Aluminium Ingot-BALCO/HINDALCO/NALCO/ VEDANTA or APTRANSCO Approved	
	Aluminium Alloy Rod /EC Grade Aluminium Metal ingots procured from open market or consignee agents shall not be used for manufacture of AAA Conductors		

27	Important packing & marking for the detailed package and marking please refer the specification	The exposed surface of the outer layer of conductor shall be wrapped with thin polythene sheet across the flange to preserve the conductor from dirt, grit and damage during transportation and handling and also to prevent ingress of rain water.